Ariel Gjaci

Postdoctoral Researcher | Multimodal Generative Learning | Spatial Audio | Human Motion

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Website | LinkedIn | GitHub | Google Scholar

Postdoctoral researcher in artificial intelligence and robotics, with a background in biomedical engineering. Research spans multimodal learning, generative models, human motion, and spatial audio, from dataset creation and model design to training and evaluation.

TECHNICAL SKILLS

Programming

Python; C/C++; R; MATLAB/Simulink; LaTeX

Machine Learning

PyTorch; Scikit-learn; TensorFlow

Training and Tools

Single-GPU training; Data parallelism; Git; SLURM

Systems

Linux; ROS 1; Windows

EDUCATION

Ph.D. Artificial Intelligence and Robotics

University of Genoa

Nov 2021 - May 2025

Research collaboration with King's College London.

M.Sc. Robotics Engineering

University of Genoa

2018 - 2021

B.Sc. Biomedical Engineering

University of Genoa

2014 - 2018

RESEARCH FOCUS

3D audio-visual learning; Human motion generation; Multimodal generative learning; Domain generalization.

LANGUAGES

Italian (native)

Albanian (native)

English (fluent)

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher

Italian Institute of Technology (IIT)

Feb 2025 - Present

- Developing few-shot room impulse response generation with counterfactual training for reverberation control (ongoing research).
- Built Eigenmike 64 processing tools for MVDR, DAS, and MaxRE beamforming, binaural audio, and acoustic energy mapping on 3D room point clouds.
- Implemented audio/video synchronization and 360-degree acoustic overlays for spatial-audio recordings.

PhD Researcher

University of Genoa

Nov 2021 - May 2025

- Designed ALaDiT, a multimodal diffusion-transformer for co-speech motion generation, within SICAGE (ECCV 2026, main conference).
- Built and released TED4C-L: 106.45 hours of synchronized motion, audio, and text features from 764 speakers across four cultural groups.
- Learned speaker-independent cultural representations with Fishr and adversarial domain generalization, evaluated on speakers excluded from training.

Software Engineer

AKKA Technologies

May 2021 - Oct 2021

- Developed Python backend software for aircraft panels and presented a working demo to stakeholders.

SELECTED PUBLICATIONS

SICAGE: Speaker-Independent Culture-Aware Gesture Generation using TED4C-L Dataset. European Conference on Computer Vision (ECCV), 2026.

Labeling Sentences with Symbolic and Deictic Gestures via Semantic Similarity. IEEE RO-MAN, 2024.

Towards Culture-Aware Co-Speech Gestures for Social Robots. International Journal of Social Robotics, 2022.

Full publication list

TEACHING AND SERVICE

Teaching assistant (C++, Android/Kotlin); ECCV workshop co-organizer; reviewer for AAAI, ECCV, ACM Multimedia, IROS, ICMI, and IJSR; IEEE RO-MAN 2024 session chair.